

ROUYN-NORANDA, QC, Canada

WMO: 717340

Lat: 48.2458N

Lon: 79.0342W

Elev: 1043

StdP: 14.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.9	-18.2	-31.2	1.0	-23.3	-25.4	1.4	-17.9	21.2	12.7	19.0	10.8	6.0	270	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	83.7	66.9	80.3	64.8	77.3	63.4	70.1	78.5	68.2	75.5	66.3	73.3	10.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.4	104.8	73.5	65.6	98.1	72.1	63.6	91.5	70.2	34.8	78.2	33.2	75.6	31.6	73.4	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	17.6	15.7	DB	-29.9	89.3	5.7	3.5	-34.0	91.8	-37.4	93.8	-40.6	95.8	-44.8	98.3	
			WB	-30.1	74.0	5.7	2.7	-34.2	76.0	-37.5	77.6	-40.7	79.1	-44.8	81.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	37.2	4.5	8.5	20.4	34.8	50.6	60.6	64.8	62.3	54.7	42.0	27.7	14.0	(6)
(7)		DBStd	23.27	14.13	12.56	12.61	10.18	9.41	8.02	6.16	6.28	8.25	8.59	10.25	12.36	(7)
(8)		HDD50	6189	1412	1163	921	463	111	12	0	1	43	279	669	1115	(8)
(9)		HDD65	10322	1877	1583	1384	905	455	176	80	128	323	713	1118	1580	(9)
(10)		CDD50	1526	0	0	2	8	128	330	458	383	185	31	1	0	(10)
(11)		CDD65	184	0	0	0	0	7	44	73	46	14	0	0	0	(11)
(12)		CDH74	1476	0	0	0	3	106	404	563	307	91	2	0	0	(12)
(13)		CDH80	319	0	0	0	0	22	88	137	56	16	0	0	0	(13)
(14)	Wind	WSAvg	7.9	7.6	8.1	8.4	8.8	8.5	7.4	7.2	6.9	7.3	8.3	8.3	7.7	(14)
(15)	Precipitation	PrecAvg	33.1	2.1	1.5	2.0	2.2	2.6	3.0	3.6	4.0	3.3	2.8	2.2		(15)
(16)		PrecMax	40.0	3.7	3.3	3.5	4.3	5.0	5.4	7.6	6.0	8.1	5.5	5.1	3.9	(16)
(17)		PrecMin	28.0	1.0	0.5	0.4	0.7	0.9	1.5	1.8	1.6	1.3	1.9	1.2		(17)
(18)		PrecStd	3.2	0.7	0.7	0.9	1.0	1.0	1.1	1.7	1.0	1.4	1.2	0.8	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	42.3	56.0	69.4	83.2	86.9	88.9	85.5	82.0	71.0	56.3	41.8	(19)
(20)			MCWB	39.0	38.7	46.2	52.6	62.2	69.4	69.5	68.1	66.7	59.6	51.2	39.4	(20)
(21)		2%	DB	34.2	36.1	47.7	60.9	76.4	82.5	83.8	80.8	76.1	64.9	49.7	36.9	(21)
(22)			MCWB	33.4	33.0	40.2	46.5	58.5	65.4	67.3	66.2	65.2	56.6	46.1	35.5	(22)
(23)		5%	DB	30.6	31.9	42.8	55.2	72.0	79.0	79.9	77.4	72.1	59.8	45.3	34.1	(23)
(24)			MCWB	29.6	30.2	36.7	43.6	56.0	62.5	65.2	64.7	62.7	54.1	42.5	32.9	(24)
(25)		10%	DB	26.5	27.8	38.1	50.2	67.2	75.0	76.6	73.8	68.3	55.4	41.1	31.8	(25)
(26)			MCWB	25.4	26.0	33.3	41.1	54.1	61.2	63.8	63.2	61.2	50.7	38.7	31.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	40.0	51.0	55.9	68.0	72.2	73.0	71.2	69.7	62.1	51.5	40.5	(27)
(28)			MCD B	39.4	41.4	52.8	66.4	75.9	82.3	82.9	79.7	76.1	68.8	54.7	41.3	(28)
(29)		2%	WB	33.3	33.3	41.6	49.2	62.5	68.8	70.2	69.1	67.1	58.1	46.8	35.6	(29)
(30)			MCD B	33.9	35.8	45.8	57.1	71.5	77.6	78.5	76.7	73.3	63.3	48.8	36.7	(30)
(31)		5%	WB	29.5	30.1	37.1	45.1	59.2	66.1	68.2	67.0	64.6	54.9	42.7	33.0	(31)
(32)			MCD B	30.4	32.0	41.9	52.5	67.5	73.7	75.4	73.8	70.5	59.0	45.1	33.9	(32)
(33)		10%	WB	25.3	26.0	33.6	41.7	55.5	63.7	66.1	64.9	61.8	51.3	38.8	30.9	(33)
(34)			MCD B	26.4	27.7	37.3	49.5	64.6	70.6	73.3	71.3	67.3	55.4	40.8	31.8	(34)
(35)	Mean Daily Temperature Range	MDBR	17.1	18.8	19.2	18.1	20.1	19.4	17.9	17.1	16.6	13.0	11.3	13.8	(35)	
(36)		5% DB	MCD BR	16.2	18.3	20.9	25.5	26.5	24.1	22.2	21.4	21.2	20.2	15.9	12.0	(36)
(37)			MCWBR	16.0	16.1	14.7	15.0	12.8	10.9	9.6	9.8	11.5	13.4	13.1	11.5	(37)
(38)		5% WB	MCD BR	16.5	18.1	18.9	22.6	22.2	20.1	18.7	18.2	18.5	18.3	15.0	12.0	(38)
(39)			MCWBR	16.3	16.4	14.6	15.1	13.4	11.8	10.0	9.9	11.4	14.0	13.4	11.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.260	0.277	0.311	0.347	0.359	0.379	0.390	0.381	0.364	0.323	0.296	0.259	(40)	
(41)		taud	2.273	2.228	2.209	2.318	2.402	2.362	2.366	2.384	2.415	2.518	2.456	2.338	(41)	
(42)		Ebn at Noon	259	279	286	286	286	280	275	272	266	263	245	245	(42)	
(43)		Edh at Noon	23	31	37	37	36	37	37	35	30	23	20	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	391	736	1187	1466	1651	1800	1724	1446	1084	637	330	265	(44)	
(45)		RadStd	61	106	122	122	147	114	146	98	102	88	38	34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page